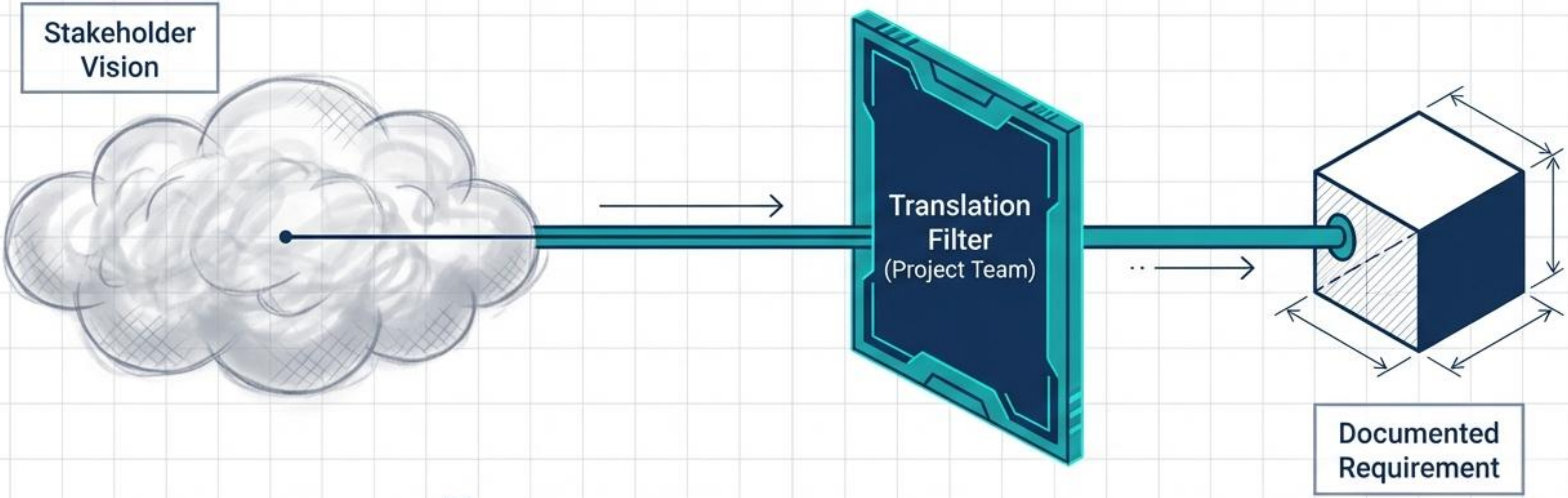
Certyfikat PMP: Project Requirements

Mastering the Tools, Techniques, and Artifacts of the Collect Requirements Process

AUTHOR: DR ZBIGNIEW GALAR

THE CHALLENGE:
Stakeholders almost always know what they want the end product to look like, but struggle to articulate their specific needs.

THE SOLUTION:
Gathering requirements is not for the faint of heart. It demands reading between the lines and asking probing questions.



KEY TAKEAWAY:
Expert communicators are required to draw the true information out of the stakeholder.

Card 1

THE EXPERT COMMUNICATOR
(Project Team)

- Reads between the lines and asks spontaneous, probing questions.
- Documents rigorously.
- Translates abstract needs into measurable project requirements.

Card 2

THE BUSINESS PROCESS OWNER
(Subject Matter Expert)

- Mid-level and line managers with their fingers on the day-to-day pulse.
- Subject matter experts for their specific area.

ANALOGY: THE BREWERY

It takes many experts to market a great beer. You must interview the machinists (equipment regulation), chemists (formula adjustment), and graphic artists (label design) to capture the full scope of requirements.

CONVERSATIONAL & GROUP (Interactive)	OBSERVATIONAL & EMPIRICAL (Action-Based)	ANALYTICAL & CREATIVE (Data & Logic)
Interviews	Observations (Job Shadowing)	Benchmarking
Focus Groups	Prototypes	Context Diagrams
Facilitated Workshops	Questionnaires and Surveys	Document Analysis
		Idea / Mind Mapping
		Group Decision-Making

INTERVIEWS

INTERVIEWS
(Depth)

- **FORMAT:** One-on-one conversations.
- **TARGET:** Subject Matter Experts (SMEs) and experienced participants.
- **ADVANTAGE:** Generates rich, detailed qualitative data. Allows for spontaneous, follow-up questions as they occur.

QUESTIONNAIRES & SURVEYS

QUESTIONNAIRES & SURVEYS
(Breadth)

- **FORMAT:** One-to-many queries.
- **TARGET:** Large groups of participants.
- **ADVANTAGE:** Gathers information rapidly. Enables precise statistical analysis of quantitative results.

EXAM SPOTLIGHT

FOCUS GROUPS

PARTICIPANTS: Gatherings of **PREQUALIFIED** subject matter experts and specific stakeholders.

LEADERSHIP: Conducted by a trained moderator.

GOAL: Deep dive into specific expectations and attitudes regarding a product or service.

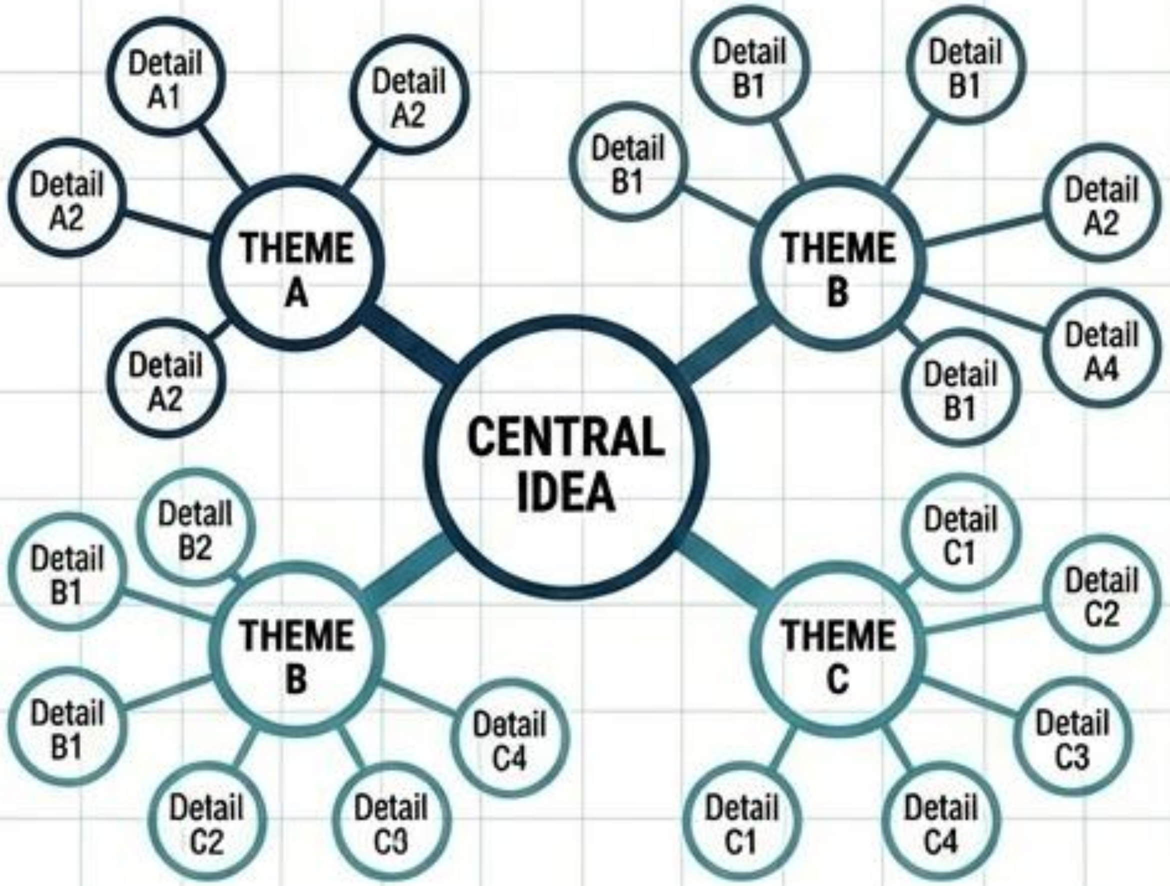
FACILITATED WORKSHOPS

PARTICIPANTS: **CROSS-FUNCTIONAL** stakeholders representing multiple different departments.

METHODOLOGIES: Joint Application Design/Development (JAD) in IT; Quality Function Deployment (QFD) in manufacturing.

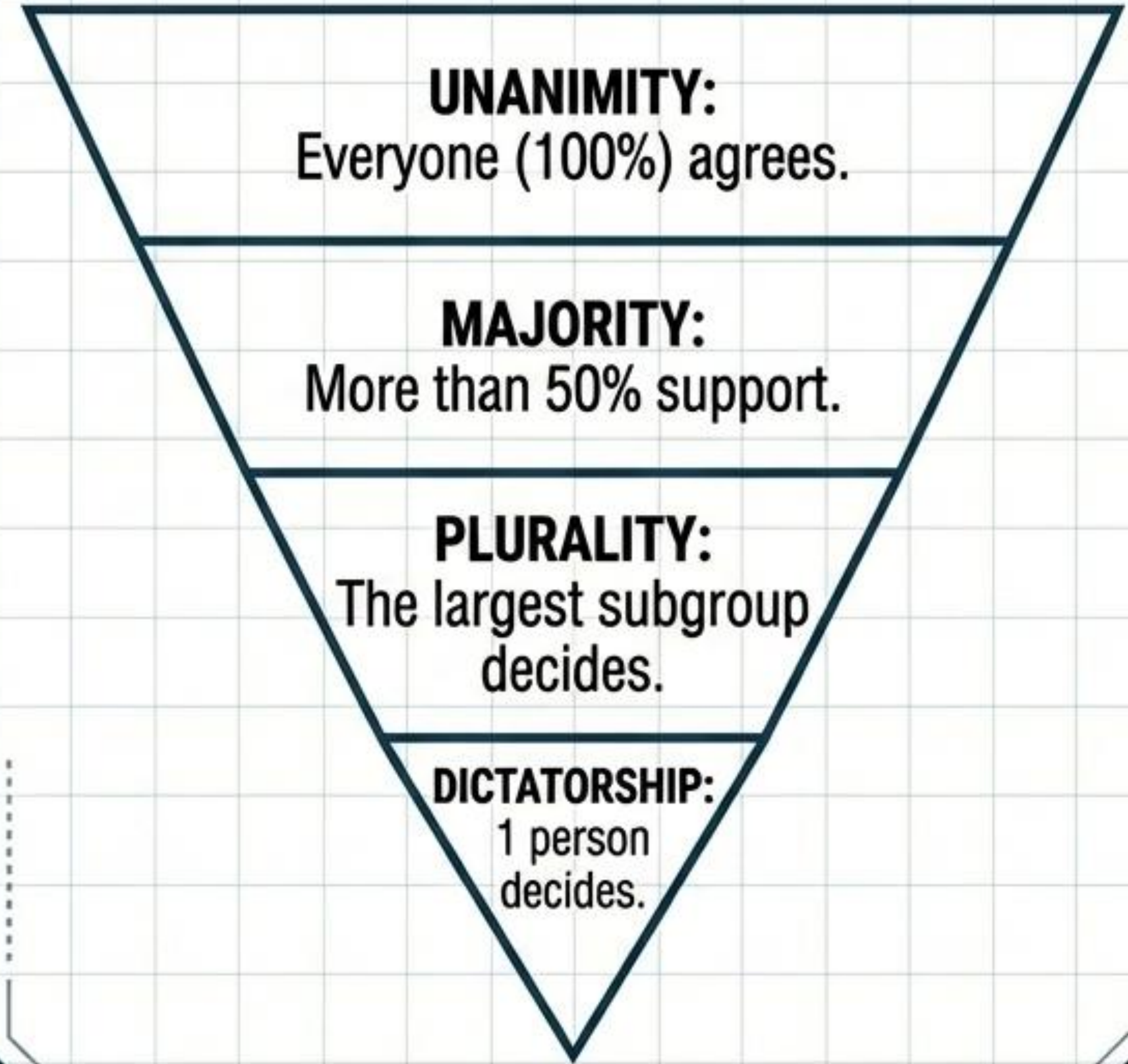
GOAL: Resolve inter-departmental differences quickly and achieve broader consensus.

IDEA / MIND MAPPING



- Participants brainstorm to record ideas.
- The facilitator groups similar topics together to understand common themes, map differences, and spark new ideas.

REACHING CONSENSUS (4 MODELS)



OBSERVATION & PROTOTYPING TECHNIQUES

OBSERVATIONS (JOB SHADOWING)

METHOD: Sitting side-by-side with users or participating directly in the work.

VALUE: Uncovers critical hidden requirements that users would not ordinarily think to articulate during a standard interview.

PROTOTYPING

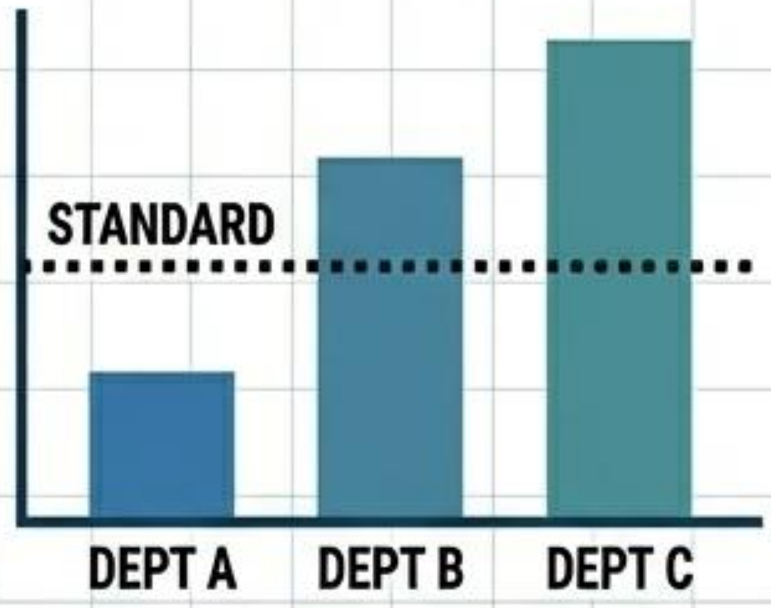
METHOD: Constructing a working model or mock-up of the final product.

VALUE: Allows participants to experiment directly with functionality, providing tangible feedback in a continuous revision cycle before final development.

CONTEXT DIAGRAMS

Visually maps interactions between people, processes, and the outputs produced.

BENCHMARKING



Comparing project metrics, processes, or operations against other departments or industry standards to promote best practices.

DOCUMENT ANALYSIS

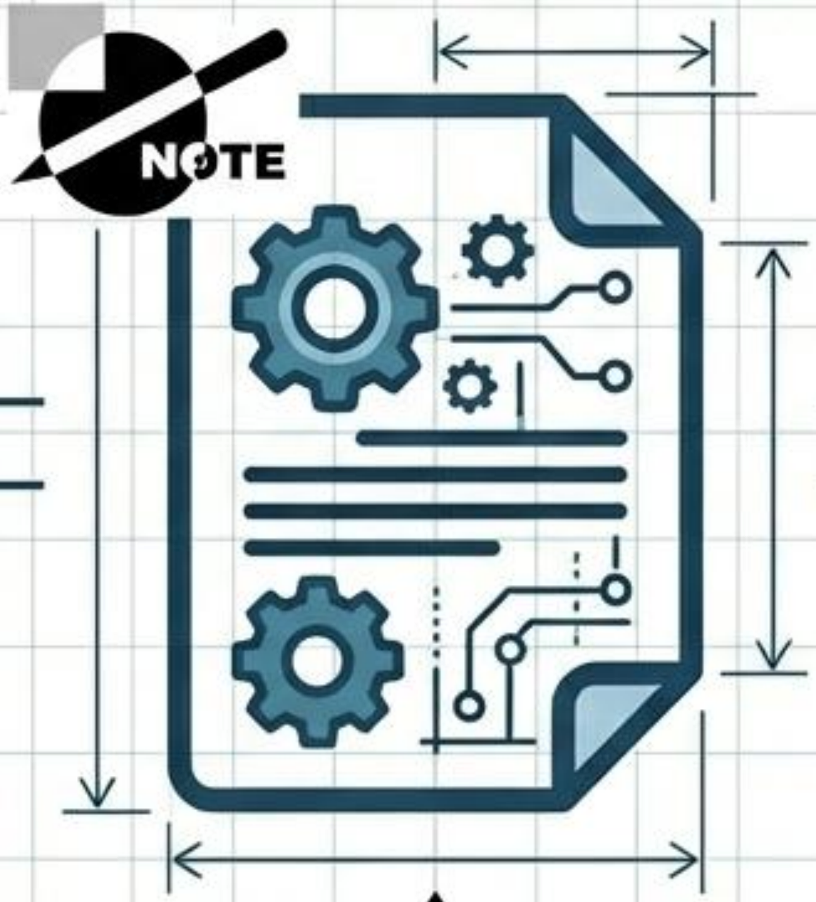
Mining existing business plans, marketing strategies, contracts, and strategic rules to extract structural constraints and requirements.

CORE PRINCIPLE: REQUIREMENTS TRACEABILITY & MEASUREMENT

Requirements must be tracked, measured, tested, and traced.
Unmeasurable requirements leave success entirely up to subjective opinions.

Business Need & Project Objectives

Transition Requirements (for operations receiving the result)



Acceptance Criteria & Quality Requirements

Assumptions & Constraints

SIGNATURES REQUIRED: Key stakeholders must sign to indicate formal acceptance of these documented requirements.

FUNCTIONAL REQUIREMENTS

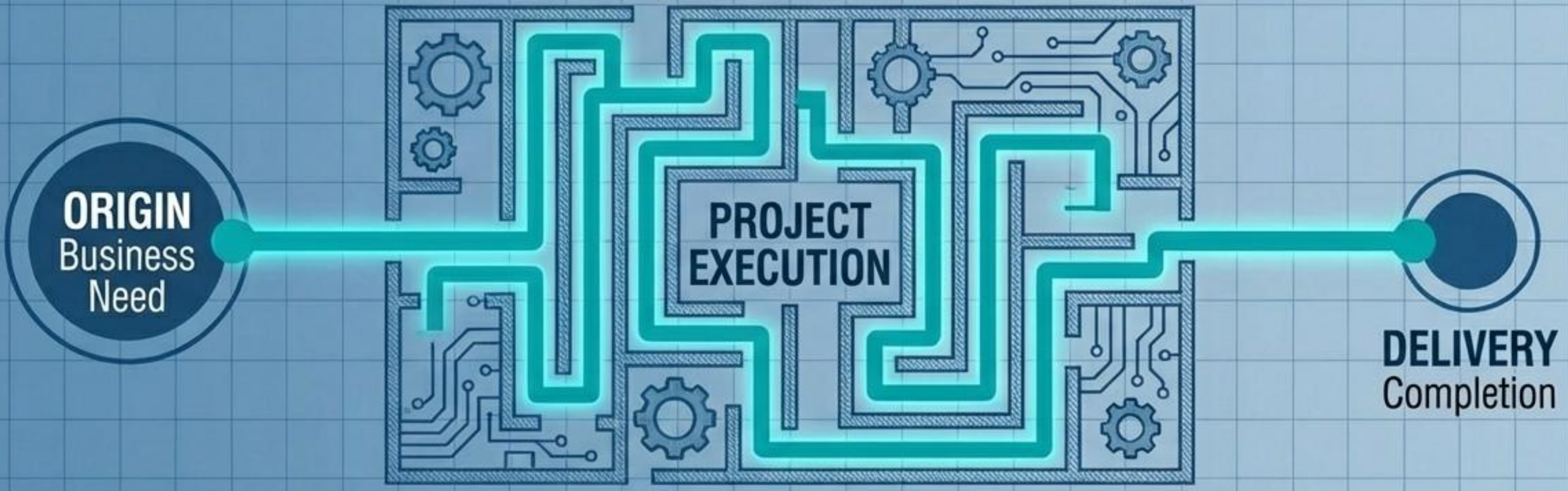
- **DEFINITION:** Describes a specific behavior or action. What the product DOES or IS.
- **SOFTWARE EXAMPLE:** Calculations or data processing rules.
- **NON-SOFTWARE EXAMPLE:** Physical specifications, exact quantities, or specific colors.

NONFUNCTIONAL REQUIREMENTS

- **DEFINITION:** Describes conditions, criteria, or constraints on how the product operates.
- **SOFTWARE EXAMPLE:** Security protocols, server uptime, or performance speed criteria.
- **NON-SOFTWARE EXAMPLE:** Environmental tolerances, regulatory compliance, or aesthetic guidelines.

THE REQUIREMENTS TRACEABILITY MATRIX (RTM):

Ensures no requirement is orphaned. It **tracks every requirement from its** original inception directly through to final testing and delivery.



EXAM SPOTLIGHT: The ultimate goal of the Traceability Matrix is to assure BUSINESS VALUE is realized. It does this by explicitly linking every single requirement back to a core business or project objective.

REQUIREMENTS TRACKING TABLE BREAKDOWN

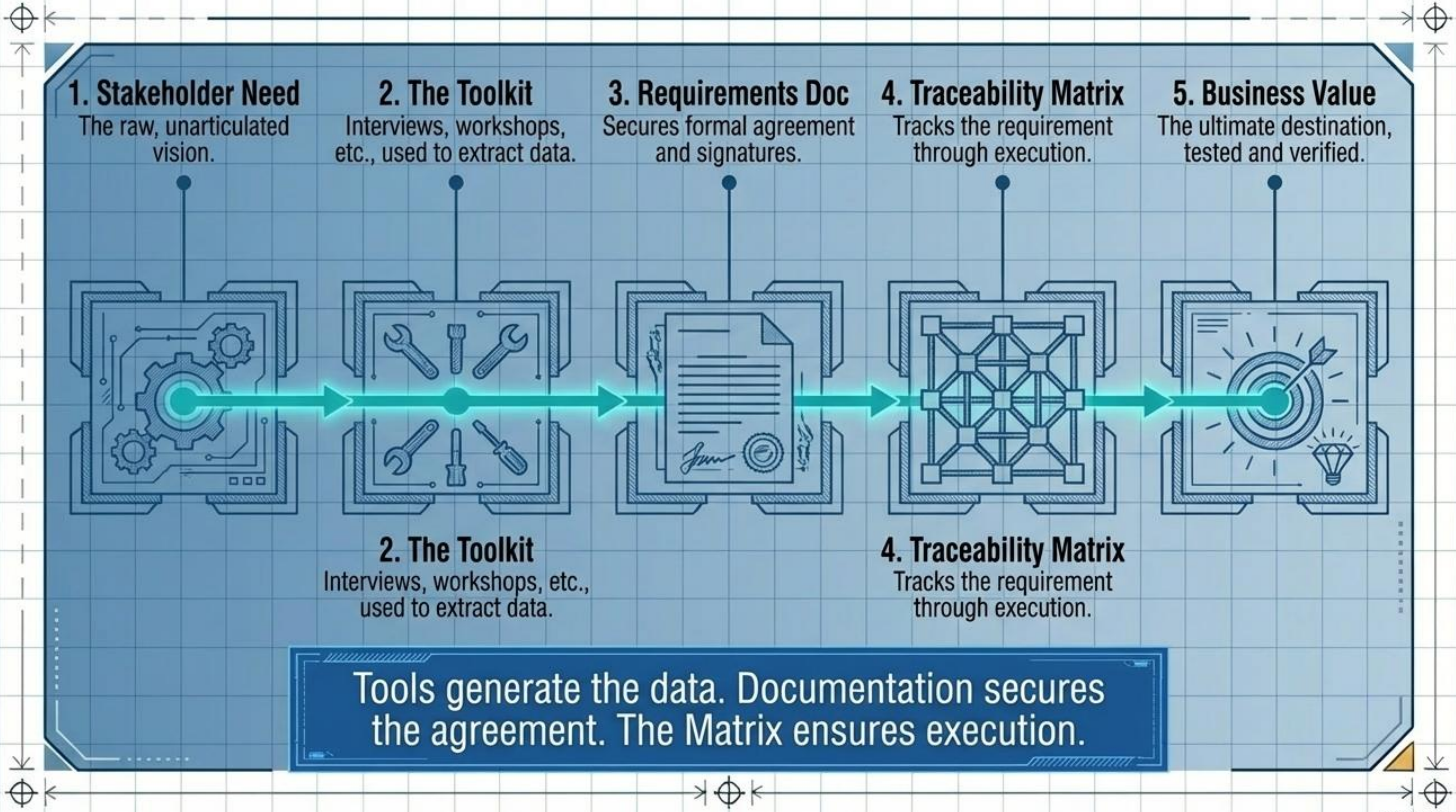
Categorizes and easily identifies the specific requirement.

Unique ID	Description of requirement	Source	Priority	Test scenario	Owner	Status
001	Requirement one	Project objective	B	User acceptance	HR specialist	Approved

Prevents scope creep by proving exactly where the requirement originated (e.g., Project Objective).

Manages constraints and trade-offs (e.g., A = Essential, B = Highly Desirable).

Defines "Done." Identifies the exact testing phase and the specific person authorized to pass or fail it.


1. Stakeholder Need
The raw, unarticulated vision.
2. The Toolkit
Interviews, workshops, etc., used to extract data.
3. Requirements Doc
Secures formal agreement and signatures.
4. Traceability Matrix
Tracks the requirement through execution.
5. Business Value
The ultimate destination, tested and verified.
2. The Toolkit
Interviews, workshops, etc., used to extract data.
4. Traceability Matrix
Tracks the requirement through execution.
Tools generate the data. Documentation secures the agreement. The Matrix ensures execution.

CURRENT STATE:
Requirements dictate the wants, needs, and expectations of the stakeholders.

NEXT STEP:
DEFINE SCOPE.
We must now translate these documented requirements into the finalized product scope (features and characteristics) the strict project scope.